

Peng Jiang

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Education

Tsinghua University	School of Life Sciences	B.Sc. in Biological Sciences	2017.08 – 2021.06
Tsinghua University	Department of Computer Science and Technology	Minor in Computer Applications	2018.08 – 2021.06
Tsinghua University	School of Life Sciences	PhD Student in Computational Neuroscience	2021.08 – present

Academic Experience

Tsinghua University	School of Life Sciences	Supervisor: Xiaoxuan Jia	2022.03 – 2024.09
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Comparing neural representations of brains and foundation models in multi-task learning

- Collected large-scale neural activity from different mouse visual brain areas with invasive Neuropixels probes, and performed low-dimensional representation extraction and neural dynamics analysis.
- Built recurrent neural network (RNN) models to simulate multi-area execution of multiple cognitive decision tasks and study the resulting dynamical representations.
- Fine-tuned large language models with LoRA or full-parameter updates for multi-task continual learning, studying compositional generalization across tasks.

NeuroHorizon: Long-Horizon Neural Activity Forecasting with Hierarchical Memory	2026
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Submitted to NeurIPS 2026 · Under review

- Studied long-horizon forward prediction of neural population activity.
- Built an autoregressive encoder-decoder model with event-level spike tokenization and hierarchical tail+segment memory.
- Evaluated multi-step rollout stability and generalization on motor-cortex and visual-cortex datasets.

Project Experience

Space Walk (Shenzhen) Technology Co., Ltd.	Co-Founder and Algorithm Lead (startup leave)	2024.09 – 2025.12
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Real-time audio-driven digital human based on 3DGS

- Built a multi-camera capture system and handled calibration and preprocessing, including intrinsic/extrinsic calibration and color calibration.
- Improved the character video processing pipeline by adding a sapiens-based tooth-segmentation loss in the track pipeline and cleaning / unifying data to reduce identity inconsistency and train-test distribution mismatch.
- Reworked the open-source audio2flame driving stack with flow matching and DiT to improve motion realism.
- Based on an open-source FLAME-based 3DGS model, fixed camera-scale issues in the monocular setting and improved reconstruction robustness through Gaussian opacity and density tuning.

NeuroGPT project, 4th place among 92 teams

- Led the team in designing NeuroGPT, using semantics decoded from neural activity with an LLM as the information transformation hub to control downstream applications.
- Focused on the path from neural semantic decoding to application-level control in interactive systems.